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INTERNSHIP REPORT

Lattice Boltzmann Method and its Implementation

Theoretical Limitations, Lattice Discretization, and OpenLB Workflows

SUBMITTED BY

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INTERNSHIP PERIOD: May 2026 – July 2026

August 2026

Abstract

This report presents the theoretical formulation and computational implementation of the Lattice Boltzmann Method (LBM) for fluid flow simulations. Tracing the progression from classical kinetic theory and continuum mechanics to discrete lattice gas automata, we formulate the Lattice Boltzmann Equation and its velocity space discretization across standard lattice structures ($D2Q9$ and $D3Q19$). Furthermore, parameter selection guidelines are analyzed to maintain numerical accuracy, Mach no.- dependent stability conditions, and computational efficiency. Finally, we detail the simulation framework, and hardware configurations used to run and visualize the simulation.

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1 Introduction

1.1 Background

1.1.1 Kinetic Theory:

When we talk about the microscopic scale [$l < 10^{-9}$ m], we say that the length of our system is equal to the molecular mean free path. The Kinetic theory in this scale is based on the fact that we are tracking individual molecules of fluid.

For our simplicity, we assume that we use dilute gas. But for real gases or liquids it is inappropriate, as we need to consider intermolecular forces there.

In the kinetic theory of gasses, we assume a dilute monoatomic gas. Gas molecules have random motion in all directions. Collisions between molecules are elastic. Therefore, the momentum and kinetic energy is always conserved.

For polyatomic gases, there are other ways in which these molecules store energy (rotational and vibrational energy) depending on their degree of freedom.

The kinetic theory of gases, relates the motion of the atoms to the volume, pressure, and temperature of the gas.

As the gas molecules are in random motion they cover the whole space(volume) of the container in which they are contained. For now, we ignore the collisions between the molecules and only consider the elastic collision between wall and the molecules.

Derivation of Ideal Gas Pressure from Kinetic Theory

Let us consider a cubical box of uniform edge length L containing n moles of an ideal gas. Each gas molecule has a mass m and a velocity vector $\vec{v} = (v_x, v_y, v_z)$. The total volume of the container will be:-

$$V = L^3 \quad (1)$$

We analyze the collisions of these molecules against a single wall perpendicular to the x -axis, which has an area of $A = L^2$.

Let a molecule with mass m approach the wall with a normal velocity component v_x . Assuming a perfectly elastic collision with the stationary solid wall, the parallel velocity components (v_y, v_z) remain unchanged, while the normal component exactly reverses from $+v_x$ to $-v_x$. The initial and final linear momenta along the x -direction are:-

$$p_{x,i} = mv_x, \quad p_{x,f} = -mv_x \quad (2)$$

The net change in momentum (Δp_x) experienced by the particle is:-

$$\Delta p_x = p_{x,f} - p_{x,i} = (-mv_x) - (mv_x) = -2mv_x \quad (3)$$

By conservation of momentum, the momentum delivered to the wall during a single impact is equal in magnitude and opposite in sign:-

$$\Delta p_{\text{wall}} = +2mv_x \quad (4)$$

The molecule then travels a distance of $2L$ [as distance between parallel walls is L] along the x -direction before arriving at the same wall and colliding with this specific wall again. The time interval Δt between consecutive impacts on this wall is:-

$$\Delta t = \frac{2L}{v_x} \quad (5)$$

Thus, the average rate at which momentum is transferred to the wall by this single molecule over time is:-

$$\frac{\Delta p_x}{\Delta t} = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L} \quad (6)$$

By Newton's second law ($\vec{F} = d\vec{p}/dt$), the total normal force F_x exerted on the wall is the sum of the momentum rates delivered by all N molecules in the system:-

$$F_x = \sum_{n=1}^N \frac{mv_{xn}^2}{L} = \frac{m}{L} (v_{x1}^2 + v_{x2}^2 + \cdots + v_{xN}^2) \quad (7)$$

The macroscopic pressure p is defined as this normal force divided by the wall surface area ($A = L^2$):-

$$p = \frac{F_x}{L^2} = \left(\frac{m}{L^3}\right) (v_{x1}^2 + v_{x2}^2 + \cdots + v_{xN}^2) \quad (8)$$

The total population of molecules can be expressed as $N = nN_A$, where N_A is Avogadro's number. We can replace the summation of individual squared velocities with the mean value $(v_x^2)_{\text{avg}}$:-

$$v_{x1}^2 + v_{x2}^2 + \cdots + v_{xN}^2 = N(v_x^2)_{\text{avg}} = nN_A(v_x^2)_{\text{avg}} \quad (9)$$

Substituting this relation, and knowing that the product mN_A equals the molar mass M of the gas while $L^3 = V$, the pressure equation simplifies to:-

$$p = \frac{nM(v_x^2)_{\text{avg}}}{V} \quad (10)$$

The magnitude of the total velocity squared for any given particle is given by $v^2 = v_x^2 + v_y^2 + v_z^2$. As the gas is isotropic in nature, we can say that

$$(v_x^2)_{\text{avg}} = (v_y^2)_{\text{avg}} = (v_z^2)_{\text{avg}} \quad (11)$$

The total mean square velocity $(v^2)_{\text{avg}}$ can therefore be written as:-

$$(v^2)_{\text{avg}} = 3(v_x^2)_{\text{avg}} \implies (v_x^2)_{\text{avg}} = \frac{1}{3}(v^2)_{\text{avg}} \quad (12)$$

Substituting this isotropic expression back into the pressure relation yields the final macroscopic pressure formula as:-

$$p = \frac{nM(v^2)_{\text{avg}}}{3V} \quad (13)$$

Further we can use the ideal gas equation and translational kinetic energy equation to prove that:-

$$K_{\text{avg}} = \frac{3}{2}k_B T \quad (14)$$

Where k_B is the Boltzmann constant and T is the absolute temperature.

1.1.2 Continuum theory:

In this case we move on from the microscopic scale to the macroscopic scale [$l > 10^{-6}$ m]. Instead of studying individual fluid molecules, we consider the fluid to a continuous system i.e. there are no empty spaces between any two points in the fluid. Hence our differential calculus rules are applicable here. Now consider a fluid element ΔV , we consider this element to be small in the system scale. This implies there are millions of molecules in random motion preserving the isotropy present in this fluid element. Their individual behaviour is averaged out.

We take the density of this fluid element to be ρ and volume as V . we apply simple law of conservation of mass here, and state that the rate of mass accumulation = rate of flow of mass in - rate of flow of mass out.

$$\text{Accumulation Rate} = \Delta x \Delta y \Delta z \frac{\partial \rho}{\partial t} \quad (15)$$

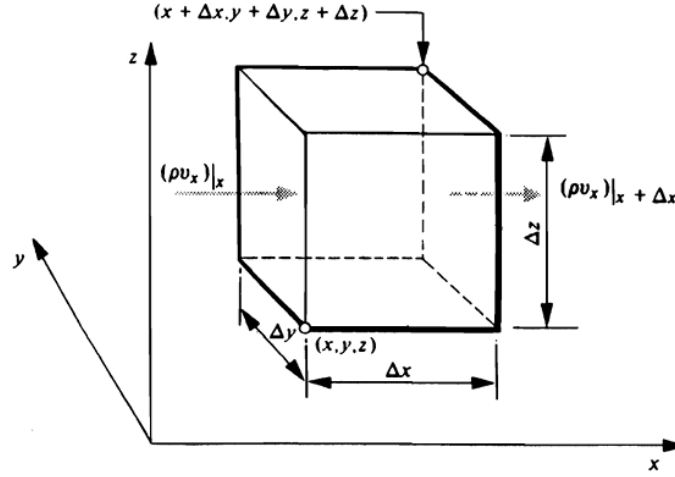


Fig. 2.4 Volume element fixed in space with fluid flowing through it.

Figure 1: Control volume for mass conservation derivation. Source: Transport Phenomena in Materials Processing: D.R. Poirier G.H. Geiger

If we calculate this change in mass flow rate in all 3 directions,

$$\begin{aligned} \Delta x \Delta y \Delta z \frac{\partial \rho}{\partial t} = & \Delta y \Delta z [\rho v_x|_x - \rho v_x|_{x+\Delta x}] \\ & + \Delta x \Delta z [\rho v_y|_y - \rho v_y|_{y+\Delta y}] \\ & + \Delta x \Delta y [\rho v_z|_z - \rho v_z|_{z+\Delta z}] \end{aligned} \quad (16)$$

Dividing Eq. (16) through by the total volume element $\Delta x \Delta y \Delta z$ yields the following formulation:

$$\frac{\partial \rho}{\partial t} = - \frac{\rho v_x|_{x+\Delta x} - \rho v_x|_x}{\Delta x} - \frac{\rho v_y|_{y+\Delta y} - \rho v_y|_y}{\Delta y} - \frac{\rho v_z|_{z+\Delta z} - \rho v_z|_z}{\Delta z} \quad (17)$$

Taking the mathematical limit as these differential spatial dimensions approach zero ($\Delta x, \Delta y, \Delta z \rightarrow 0$) results in the standard equation of continuity:

$$\frac{\partial \rho}{\partial t} = - \left[\frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z) \right] \quad (18)$$

For an incompressible fluid, $\partial \rho / \partial t = 0$ and the density scalar factors out of the spatial derivatives, reducing the continuity relation to:

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 \quad (19)$$

Using the divergence operator, the incompressible flow condition is written as:

$$\nabla \cdot \mathbf{v} = 0 \quad (20)$$

Similarly, if we apply conservation of momentum on the fluid element we can get the famous Navier-Stokes equation :-

$$dm = \rho dV = \rho dx dy dz \quad (21)$$

Applying Newton's second law of motion ($\sum \mathbf{F} = d\mathbf{m}\mathbf{a}$) to this fluid element :-

$$\sum \mathbf{F} = \rho dx dy dz \frac{D\mathbf{u}}{Dt} \quad (22)$$

where $\mathbf{u} = (u, v, w)$ is the velocity vector field and $\frac{D\mathbf{u}}{Dt}$ represents the substantial (or material) derivative tracking total fluid acceleration.

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial\mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} \quad (23)$$

The net forces acting on the fluid volume element are classified into two categories: body forces and surface forces.

Body forces act uniformly throughout the entire mass of the fluid element. Neglecting electromagnetic forces, we only consider gravitational force acting on the system.

$$\mathbf{F}_B = dm \mathbf{g} = \rho \mathbf{g} dxdydz \quad (24)$$

Surface forces arise from stresses acting across the exterior boundaries of the fluid element. These consist of isotropic fluid pressure forces which act normally in the inward direction[compressive] and viscous frictional shear forces apposing this motion.

$$\sigma_{xx} = \underbrace{-P}_{\text{always compressive}} + \tau_{xx} \quad (25)$$

To find the net surface force in any given direction, we perform a first-order Taylor series expansion across opposite faces of the differential cube.

Analyzing forces along the x -axis, three surface contributions exist: normal stresses on the x -faces, and tangential shear stresses on both the y -faces and z -faces. The net force components are calculated by subtracting the incoming face values from the exiting face values:

$$\text{Net } x\text{-Normal Force} = \left(\sigma_{xx} + \frac{\partial\sigma_{xx}}{\partial x} dx \right) dydz - \sigma_{xx} dydz = \frac{\partial\sigma_{xx}}{\partial x} dxdydz \quad (26)$$

$$\text{Net } y\text{-Shear Force in } x = \left(\sigma_{yx} + \frac{\partial\sigma_{yx}}{\partial y} dy \right) dx dz - \sigma_{yx} dx dz = \frac{\partial\sigma_{yx}}{\partial y} dxdydz \quad (27)$$

$$\text{Net } z\text{-Shear Force in } x = \left(\sigma_{zx} + \frac{\partial\sigma_{zx}}{\partial z} dz \right) dxdy - \sigma_{zx} dxdy = \frac{\partial\sigma_{zx}}{\partial z} dxdydz \quad (28)$$

Summing these terms up gives the total differential surface force acting in the x -direction ($dF_{S,x}$):

$$dF_{S,x} = \left(\frac{\partial\sigma_{xx}}{\partial x} + \frac{\partial\sigma_{yx}}{\partial y} + \frac{\partial\sigma_{zx}}{\partial z} \right) dxdydz \quad (29)$$

For a Newtonian fluid, stress is linearly proportional to the rate of fluid deformation (velocity gradients). Assuming a constant density ρ (incompressible limit) and constant dynamic viscosity μ , the total surface stresses are given as:-

$$\sigma_{xx} = -p + 2\mu \frac{\partial u}{\partial x}, \quad \sigma_{xy} = \sigma_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \quad (30)$$

$$\sigma_{yy} = -p + 2\mu \frac{\partial v}{\partial y}, \quad \sigma_{yz} = \sigma_{zy} = \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \quad (31)$$

$$\sigma_{zz} = -p + 2\mu \frac{\partial w}{\partial z}, \quad \sigma_{xz} = \sigma_{zx} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \quad (32)$$

The classic Navier-Stokes equations in cartesian coordinate components:

x -Momentum Component:

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho g_x \quad (33)$$

y-Momentum Component:

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \rho g_y \quad (34)$$

z-Momentum Component:

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + \rho g_z \quad (35)$$

In standard vector notation, the incompressible Navier–Stokes equation is written as:

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right] = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g} \quad (36)$$

Note that in this complete derivation we made the following assumptions:-

- Fluid is isotropic
- Newtonian fluid
- Stokesian fluid

1.1.3 Lattice gas models

In the previous theories we studied about the microscopic scale where we studied about how each molecule's motion contributes to macroscopic properties like Pressure, temperature. We also studied about the continuum theory where we consider a larger scale[macroscopic] to define a fluid element apply conservation laws. What if we stop in between these two scales. These lattice gas models are based on mesoscale [10^{-9} to 10^{-6} m]. In this method, the space, time and velocity are discretized. Space is discretized using grid lines and each intersection is represented by a node.

The first model on lattice gas model was developed by Hardy, Pomeau, and de Pazzis in 1973 [HPP model], where a simple 2D lattice structure was build and each node was associated with 4 velocity directions that is a particle at each node can travel to 4 neighboring nodes.

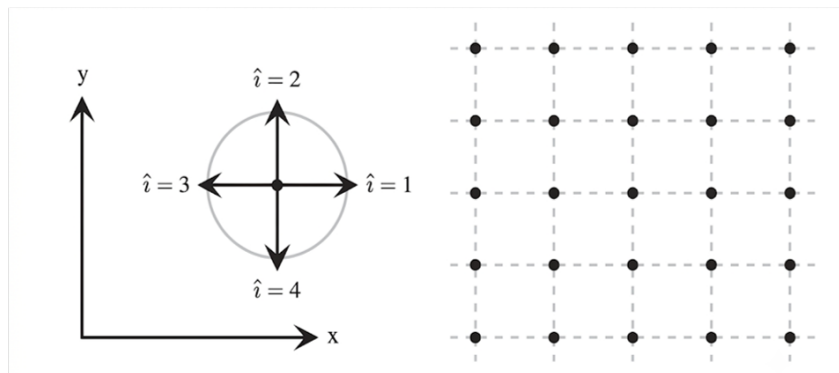


Figure 2: Lattice structure for HPP Model

We define a fictitious particle that is only the function of space, velocity and time. At each direction a particle may be present or absent depending on its occupation. The occupation is defined as $n(x, t)$. when particles collide at a node they scatter in different directions preserving

mass and momentum. In absence of collisions, the particles move one lattice unit forward in a time step. Note that each particle has same unit velocity.

Only one particle of a certain velocity (c_i) can be present at a particular node.

$$n_i(\vec{x}, t) \quad [\text{Boolean variable}]$$

$$n_i \in \{0, 1\}$$

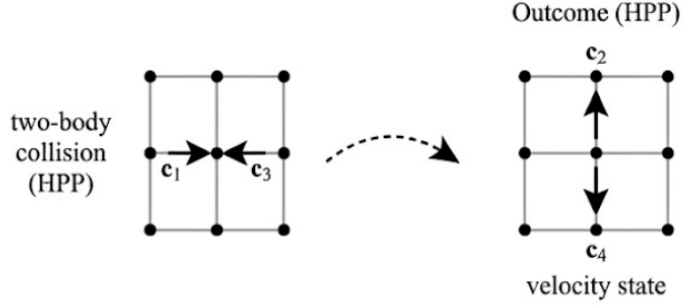


Figure 3: Collision of 2 particles at a node in HPP model

As lattice isotropy isn't maintained in this model, we have another better model named the FHP model. [However, the HPP model fails to reproduce isotropic fluid behavior. Because the square lattice only has four directions, the momentum flux tensor is anisotropic — the viscosity of the simulated fluid depends on direction, which is nonphysical for a Newtonian fluid.]

The FHP (Frisch, Hasslacher, and Pomeau [1986]) model includes 6 velocity directions instead of four and hence the particles preserve isotropy. In 2-particle collision the particles have either 2 ways in which they can propagate.

The mass and momentum equations are :-

$$\rho(x, t) = \frac{m}{v_0} \sum_i n_i(x, t), \quad \rho u(x, t) = \frac{m}{v_0} \sum_i c_i n_i(x, t), \quad (37)$$

respectively, where m is the mass of a particle and v_0 is the volume covered by the node.

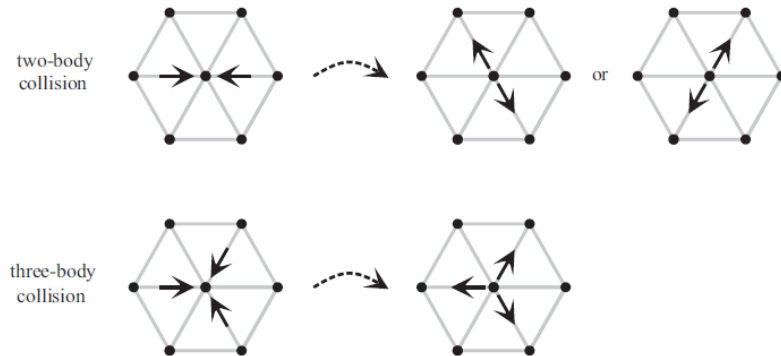


Figure 4: ref: Cellular Automata and Lattice Gases/fab.cba.mit.edu

The collision result is given by the equation :-

$$\underbrace{n_i^*(\vec{x}, t)}_{\text{post-collision}} = n_i(\vec{x}, t) + \Omega_i(\vec{x}, t)$$

$$\Omega_i \in \{0, -1, 1\} \quad \begin{cases} -1 : \text{annihilation} \\ 0 : \text{no change} \\ 1 : \text{generation} \end{cases}$$

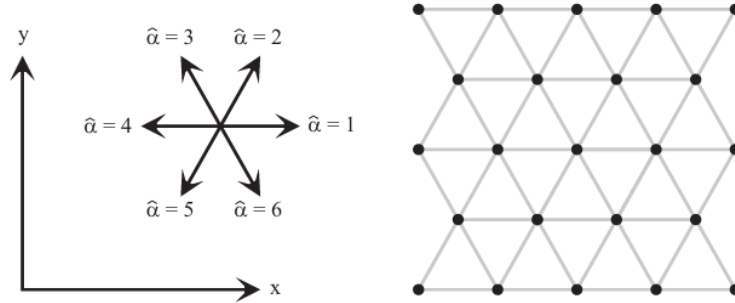


Figure 10.1. Direction indices for a triangular lattice.

Figure 5: ref: Cellular Automata and Lattice Gases/fab.cba.mit.edu

The major issue with lattice gas models is the statistical noise. These lattice gases behave like real gases at microscopic level and hence if our final goal is to simulate at macroscopic scale; the fluctuations produced at microscopic scale become useless or create chaos. The density and macroscopic velocities are simply averaged out based on the occupation $[0,1]$. These values will also fluctuate for gas at equilibrium.

The lattice gas model is the foundation of LBM, where we discretize everything. We will overcome these problems faced in lattice gases in the LBM.

1.2 Objectives

- To understand LBM derivations and fundamentals
- To analyze the stability and accuracy of these models
- To compare computation time and memory usage on different processing units(CPU vs GPU).

2 Theoretical Background

2.1 Lattice Boltzmann Method

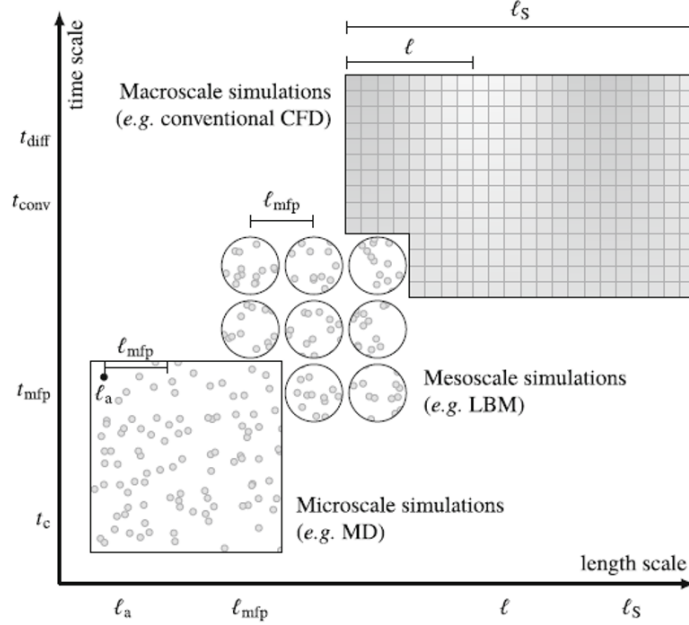


Figure 6: Ref: The Lattice Boltzmann Method- Principles and Practice – Timm Krüger, Halim Kusumaatmaja, Alexandr Kuzmin

The LBM is an updated version of Lattice gas models performed in the mesoscale. Like LG models we define grids and these particle interactions happen at nodes.

2.1.1 Lattice Boltzmann Equation and Discretization

The lattice Boltzmann equation is the principal equation of LBM. It is an updated version of lattice gas models. Instead of tracking the occupation Boolean of the particles, we calculate its probability distribution as a function of space, velocity and time.

We define,

$$f(\vec{x}, \vec{\epsilon}, t)$$

as the number of particles with vel. ϵ at location x .

If there is an external force F :-

New location = $x + \epsilon dt$; New velocity = $\epsilon + F/m dt$

If there are no collisions; the no. of particles is conserved,

$$f\left(x + \epsilon dt, \epsilon + \frac{F}{m} dt, t + dt\right) dV d^3\epsilon = f(r, \epsilon, t) dV d^3\epsilon \quad (38)$$

Now we know that not all particles leaving location x with velocity ϵ at time t will reach $x + \epsilon dt$ with velocity $\epsilon + F/m dt$. There will be collisions taking place that will lead to loss and gain of some particles and hence we define Γ_i ,

- $\Gamma_i^{(-)}$ = fractional rate at which particles between $(\epsilon, \epsilon + d\epsilon)$ and positions $(x, x + dx)$ are lost due to collision with i^{th} particle

- $\Gamma_i^{(+)}$ = fractional rate at which particles between $(\epsilon, \epsilon + d\epsilon)$ and positions $(x, x + dx)$ are added due to collision with i^{th} particle

$$\text{No. of particles lost} = \Gamma_i^{(-)} dV d^3\epsilon dt$$

$$\text{No. of particles added} = \Gamma_i^{(+)} dV d^3\epsilon dt \quad [\text{as rate}]$$

Conservation of particles:

$$f\left(x + \epsilon dt, \epsilon + \frac{F}{m} dt, t + dt\right) dV d^3\epsilon = f(x, \epsilon, t) dV d^3\epsilon + \sum_i \left(\Gamma_i^{(+)} - \Gamma_i^{(-)}\right) dV d^3\epsilon dt$$

Expanding the left-hand side using a Taylor series:

$$f(x, \epsilon, t) + \epsilon_\alpha dt \frac{\partial f}{\partial x_\alpha} + \frac{F_\alpha}{m} dt \frac{\partial f}{\partial \epsilon_\alpha} + dt \frac{\partial f}{\partial t} + \dots = f(x, \epsilon, t) + \sum_i \left(\Gamma_i^{(+)} - \Gamma_i^{(-)}\right) dt$$

Dividing through by dt gives the continuous Boltzmann transport equation:

$$\frac{\partial f}{\partial t} + \epsilon \frac{\partial f}{\partial x} + \frac{F}{m} \frac{\partial f}{\partial \epsilon} = \sum_i \left(\Gamma_i^{(+)} - \Gamma_i^{(-)}\right)$$

This leads to our LBE,

$$f_i(x + c_i \Delta t, t + \Delta t) = f_i(x, t) - \frac{\Delta t}{\tau} [f_i(x, t) - f_i^{\text{eq}}(x, t)] \quad (39)$$

Similarly conserving macroscopic density and momentum we get,

$$\rho(x, t) = \sum_i f_i(x, t), \quad \rho u(x, t) = \sum_i c_i f_i(x, t). \quad (40)$$

The collision factor used in LBE is complicated and hence is replaced by a simpler linear term named the BGK operator.

$$\Omega_i(f) = -\frac{f_i - f_i^{\text{eq}}}{\tau} \Delta t \quad (41)$$

τ = relaxation time, it is the time required for the particle to reach equilibrium.

The equilibrium is given by,

$$f_i^{\text{eq}}(\mathbf{x}, t) = \omega_i \rho \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{u^2}{2c_s^2} \right] \quad (42)$$

The D2Q9 discrete velocity vectors $\mathbf{c}_i$ in lattice units are:

$$\mathbf{c}_i \in \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right\} \quad (43)$$

The corresponding weight factors w_i are:

$$w_i = \begin{cases} \frac{4}{9} & i = 0 \quad (\text{rest}) \\ \frac{1}{9} & i = 1, 2, 3, 4 \quad (\text{axial directions}) \\ \frac{1}{36} & i = 5, 6, 7, 8 \quad (\text{diagonal directions}) \end{cases} \quad (44)$$

These weights satisfy $\sum_i w_i = 1$. The lattice speed of sound in lattice units is:

$$c_s = \frac{c}{\sqrt{3}} = \frac{\Delta x}{\sqrt{3} \Delta t}, \quad c_s^2 = \frac{1}{3} \quad (45)$$

where $c = \Delta x / \Delta t$ is the lattice speed. The D3Q19 model (Fig. 8) extends this to three dimensions with 19 velocity directions and requires more memory than D2Q9.

The famous expansion that recovers Navier Stokes equation from LBE is the Chapman-Enskog analysis [2nd order accurate].

We get the relation ,

$$\nu = c_s^2 \left(\tau - \frac{\Delta t}{2} \right), \quad (46)$$

Hence our relaxation time gets bound by this relation, which further is an important factor in parameters selection.

2.2 LBM algorithm

The overall LBM algorithm is given by these following steps:-

1. Initialize the distribution function,

$$f_\alpha = f_\alpha^{\text{eq}} [\rho(\vec{x}, 0), \vec{u}(\vec{x}, 0)] \quad (47)$$

2. Compute macroscopic moments from these values,

$$\rho(x, t) = \sum_i f_i(x, t), \quad \rho u(x, t) = \sum_i c_i f_i(x, t).$$

3. Compute the equilibrium distribution f_i^{eq} from ρ and $\mathbf{u}$.
4. Collision step.
5. Streaming step.
6. Apply the boundary conditions.
7. Go back to **step 2** until the loop is completed.

2.3 Lattice Structure

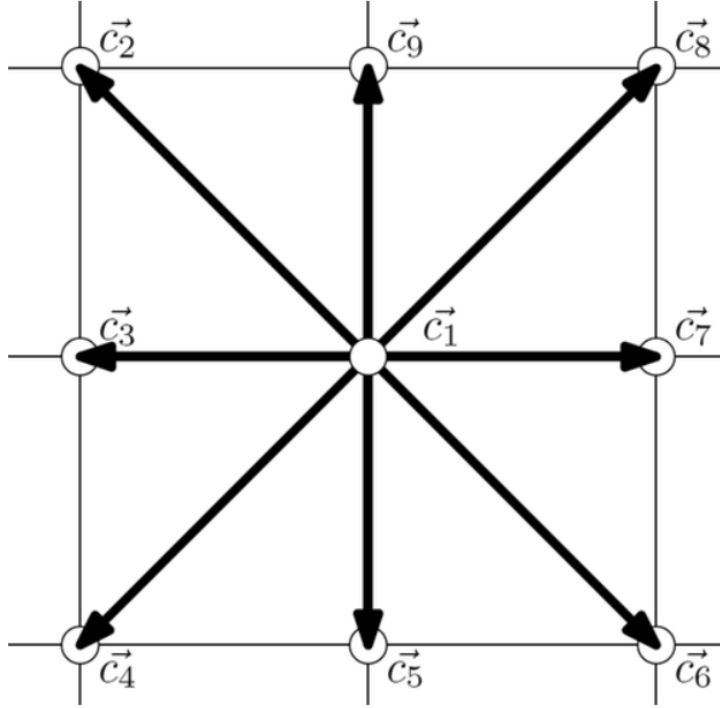


Figure 7: D2Q9 (2D)

Fig. 7 is D2Q9 lattice structure. This structure is widely used for 2D simulations. This requires less memory.

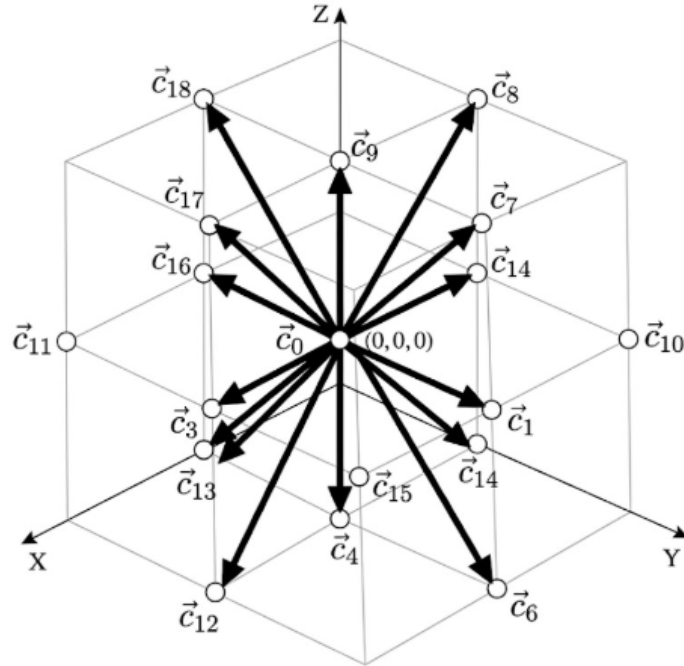


Figure 8: D3Q19 (3D)

Fig. 8 is D3Q19 lattice structure. This structure is used for 3D simulations. This requires

more memory as compared to D2Q9

3 Non-Dimensionalisation

Non-Dimensionalisation of quantities is done to make the calculations more trivial. This is achieved by dividing a dimensional quantity by the reference quantity of the same dimensions. The reference quantity is known as the conversion factor.

The key conversion relations between physical (dimensional) and lattice (dimensionless) quantities are:

$$x^* = \frac{x}{\Delta x} \quad (48)$$

$$t^* = \frac{t}{\Delta t} \quad (49)$$

where starred quantities are dimensionless lattice-unit values. Therefore, any dimensional parameter can be represented in lattice form by dividing by its corresponding conversion factor. Fig. 9 shows an example of the UnitConverter output from OpenLB for a 2D Poiseuille flow simulation, listing all conversion factors automatically.

```
[UnitConverter] ----- UnitConverter information -----
[UnitConverter] -- Parameters:
[UnitConverter] Resolution:                N=                50
[UnitConverter] Lattice velocity:          latticeU=         0.00666667
[UnitConverter] Lattice relaxation frequency: omega=         1.66667
[UnitConverter] Lattice relaxation time:    tau=             0.6
[UnitConverter] Characteristical length(m): charL=           1
[UnitConverter] Characteristical speed(m/s): charU=           1
[UnitConverter] Phys. kinematic viscosity(m^2/s): charNu=      0.1
[UnitConverter] Phys. density(kg/m^d):      charRho=          1
[UnitConverter] Characteristical pressure(N/m^2): charPressure= 0
[UnitConverter] Mach number:                machNumber=      0.011547
[UnitConverter] Reynolds number:            reynoldsNumber= 10
[UnitConverter] Knudsen number:             knudsenNumber=    0.0011547
[UnitConverter] Characteristical CFL number: charCFLnumber=    0.00666667
[UnitConverter]
[UnitConverter] -- Conversion factors:
[UnitConverter] Voxel length(m):            physDeltaX=      0.02
[UnitConverter] Time step(s):               physDeltaT=      0.000133333
[UnitConverter] Velocity factor(m/s):       physVelocity=    150
[UnitConverter] Density factor(kg/m^3):     physDensity=     1
[UnitConverter] Mass factor(kg):            physMass=      8e-06
[UnitConverter] Viscosity factor(m^2/s):    physViscosity=   3
[UnitConverter] Force factor(N):            physForce=     9
[UnitConverter] Pressure factor(N/m^2):     physPressure=   22500
[UnitConverter] -----
```

Figure 9: UnitConverter information displayed when we run a simulation a simple Poiseuille flow[2D] using OpenLB

4 Parameter Selection

4.1 Accuracy

Accuracy of the model depends heavily on N(resolution) of the model. As the velocity is second order accurate in LB. Therefore, as we increase the resolution, Δx decreases and hence the velocity errors decrease proportionally to (Δx^2) .

```

[UnitConverter] -- Parameters:
[UnitConverter] Resolution:           N=           50
[UnitConverter] Lattice velocity:      latticeU=      0.00666667
[UnitConverter] Lattice relaxation frequency: omega=      1.66667
[UnitConverter] Lattice relaxation time: tau=          0.6
[UnitConverter] Characteristical length(m): charL=        1
[UnitConverter] Characteristical speed(m/s): charU=        1
[UnitConverter] Phys. kinematic viscosity(m^2/s): charNu=    0.1
[UnitConverter] Phys. density(kg/m^d): charRho=          1
[UnitConverter] Characteristical pressure(N/m^2): charPressure= 0
[UnitConverter] Mach number:           machNumber=    0.011547
[UnitConverter] Reynolds number:        reynoldsNumber= 10
[UnitConverter] Knudsen number:         knudsenNumber=  0.0011547
[UnitConverter] Characteristical CFL number: charCFLnumber= 0.00666667

```

Figure 10: Resolution=50

Now if we increase N to 100,

```

[UnitConverter] -- Parameters:
[UnitConverter] Resolution:           N=          100
[UnitConverter] Lattice velocity:      latticeU=     0.00333333
[UnitConverter] Lattice relaxation frequency: omega=     1.66667
[UnitConverter] Lattice relaxation time: tau=           0.6
[UnitConverter] Characteristical length(m): charL=        1
[UnitConverter] Characteristical speed(m/s): charU=        1
[UnitConverter] Phys. kinematic viscosity(m^2/s): charNu=    0.1
[UnitConverter] Phys. density(kg/m^d): charRho=          1
[UnitConverter] Characteristical pressure(N/m^2): charPressure= 0
[UnitConverter] Mach number:           machNumber=    0.0057735
[UnitConverter] Reynolds number:        reynoldsNumber= 10
[UnitConverter] Knudsen number:         knudsenNumber=  0.00057735
[UnitConverter] Characteristical CFL number: charCFLnumber= 0.00333333

```

Figure 11: Resolution=100

(Δx) decreases and therefore velocity error.

Note that this comes at the cost of memory and computation time (refer efficiency).

4.2 Stability

From the Chapman-Enskog expansion we know that,

$$\nu = c_s^2 \left(\tau - \frac{\Delta t}{2} \right),$$

Figure 12: Relation of Kinematic viscosity(ν) with relaxation time(τ)

As we know that kinematic viscosity can never be negative, therefore the relaxation time (τ) has to be greater than 0.5.

The other major criteria that $Ma < 0.3$.

This can be studied by another simply derived formula for N,

$$\begin{aligned}
u_{\max}^* &< 0.1732 \\
u_{\max} \times \frac{\Delta t}{\Delta x} &< 0.1732 \\
u_{\max} \times \frac{1 \times (\tau^* - 0.5)}{3 \times \nu} \times \frac{L_c}{N} &< 0.1732 \\
N &> \frac{u_{\max}(\tau^* - 0.5)L_c}{\nu \times 0.5196} \\
\boxed{N &> \frac{\text{Re}(\tau^* - 0.5)}{0.5196}} & (50)
\end{aligned}$$

where:

- $u_{\max}^*$ = Maximum dimensionless lattice velocity (lattice Mach limit)
- $u_{\max}$ = Maximum physical fluid velocity (m/s)
- Δt = Discrete time step (s)
- Δx = Lattice grid spacing (m)
- τ^* = Dimensionless relaxation time parameter
- ν = Kinematic viscosity of the fluid (m^2/s)
- L_c = Characteristic physical length scale (m)
- N = Number of lattice grid cells along the characteristic length ($N = L_c/\Delta x$)
- Re = Reynolds number ($\text{Re} = \frac{u_{\max} L_c}{\nu}$)

Taking the previous example of Poiseuille flow,

```

[prepareLattice] Prepare Lattice ...
[UnitConverter] ----- UnitConverter information -----
[UnitConverter] -- Parameters:
[UnitConverter] Resolution:                               N=                10
[UnitConverter] Lattice velocity:                         latticeU=          0.666667
[UnitConverter] Lattice relaxation frequency:              omega=          1.666667
[UnitConverter] Lattice relaxation time:                  tau=              0.6
[UnitConverter] Characteristical length(m):               charL=              1
[UnitConverter] Characteristical speed(m/s):              charU=              1
[UnitConverter] Phys. kinematic viscosity(m^2/s):         charNu=           0.005
[UnitConverter] Phys. density(kg/m^3):                   charRho=              1
[UnitConverter] Characteristical pressure(N/m^2):         charPressure=        0
[UnitConverter] Mach number:                             machNumber=        1.1547
[UnitConverter] Reynolds number:                         reynoldsNumber=    200
[UnitConverter] Knudsen number:                         knudsenNumber=    0.0057735
[UnitConverter] Characteristical CFL number:             charCFLnumber=    0.666667
[UnitConverter]
[UnitConverter] -- Conversion factors:
[UnitConverter] Voxel length(m):                         physDeltaX=         0.1
[UnitConverter] Time step(s):                             physDeltaT=        0.0666667
[UnitConverter] Velocity factor(m/s):                   physVelocity=        1.5
[UnitConverter] Density factor(kg/m^3):                 physDensity=         1
[UnitConverter] Mass factor(kg):                         physMass=          0.001
[UnitConverter] Viscosity factor(m^2/s):                 physViscosity=       0.15
[UnitConverter] Force factor(N):                         physForce=          0.0225
[UnitConverter] Pressure factor(N/m^2):                  physPressure=        2.25
[UnitConverter] -----

```

Figure 13: $\text{Ma} \gg 0.3$

Here, the Ma no. is way higher than 0.3 when we run this simulation we get an error,

Hence it is necessary to select the resolution of the model according to the Reynolds no. requirements.

```
[error] pressure-L2-error(abs)=2.31256e+12; pressure-L2-error(rel)=3.8776e+13
[error] pressure-Linf-error(abs)=1.96122e+13; pressure-Linf-error(rel)=2.58055e+14
terminate called after throwing an instance of 'std::runtime_error'
what(): Simulation diverged.
Aborted (core dumped)
```

Figure 14: Stability errors in OpenLB model

4.3 Efficiency

Efficiency of a model is based on two main factors- Computation time, Memory required.

One of the major advantages of LBM over convention CFD is its capability of parallel computing.

OpenMP speeds up a CPU by dividing the mesh across processor cores (usually 8 to 16), which still forces each core to process thousands of cells one after another. In contrast, the GPU uses thousands of dedicated cores to update nearly every single grid cell at the exact same moment. Thus, saving computational time. Note that we are using NVIDIA's CUDA toolkit here for our GPU requirements.

Here is an example comparing the MLUPS(Million Lattice Site Updates Per Second) and run time of the same simulation on CPU and GPU,

```
[Timer]
[Timer] -----Summary:Timer-----
[Timer] measured time (rt) : 325.109s
[Timer] measured time (cpu): 1239.972s
[Timer] average MLUPs :      157.761
[Timer] average MLUPps:      39.440
[Timer] -----
atharva@DESKTOP-UGR8UHG:~/olb_cpu/examples/laminar/poiseuille2d$ |
```

Figure 15: CPU (OpenMP)

```
[Timer]
[Timer] -----Summary:Timer-----
[Timer] measured time (rt) : 95.521s
[Timer] measured time (cpu): 95.831s
[Timer] average MLUPs :      536.945
[Timer] average MLUPps:      536.945
[Timer] -----
atharva@DESKTOP-UGR8UHG:~/olb_gpu/examples/laminar/poiseuille2d$ paraview
```

Figure 16: GPU (NVIDIA CUDA Toolkit)

5 Conclusion

From Chapman-Enskog expansion, kinematic viscosity dictates that the dimensionless relaxation time must satisfy $\tau > 0.5$ to maintain positive physical viscosity and prevent mathematical breakdown.

Furthermore, the equilibrium distribution f_i^{eq} is derived from a second-order Taylor expansion of the Maxwell-Boltzmann distribution in $\mathbf{u}/c_s$. The truncated higher-order terms grow as $\mathcal{O}(\text{Ma}^3)$, so the Mach number must satisfy $\text{Ma} < 0.3$ for physically accurate results. This imposes a minimum resolution requirement:

$$N > \frac{\text{Re}(\tau^* - 0.5)}{0.5196} \quad (51)$$

which must be verified before trusting any LBM result.

If the parameters are selected wisely, LBM is a very good tool to simulate fluid flow (complex boundary conditions) efficiently.

Table 7.2 Overview of intrinsic LB limitations and suggested remedies

Accuracy	(i) τ^* not much larger than unity (ii) $U_{\max}^* < 0.03\text{--}0.1$ (iii) Δx sufficiently fine to resolve all flow features (iv) Δt sufficiently fine to reduce time discretisation artefacts
Stability	(i) $U_{\max}^* < 0.4$ (ii) τ^* not too close to $1/2$, cf. (7.18) (iii) Sufficiently small grid Reynolds number, cf. (7.19)
Efficiency	Lattice constant Δx and time step Δt not unnecessarily small (memory requirements $\propto \Delta x^{-d}$, simulation runtime $\propto \Delta t^{-1} \Delta x^{-d}$ in d dimensions)

Figure 17: Ref: The Lattice Boltzmann Method- Principles and Practice – Timm Krüger, Halim Kusumaatmaja, Alexandr Kuzmin

6 Future Work

Extend the current Lattice Boltzmann framework to model Diffusion-Limited Aggregation (DLA) phenomena.

Introducing multiphase models like Shan-Chen or Phase-Field models. Using MRT operator instead of BGK operator.

7 Simulation Framework and Hardware Specifications

Category	Tool / Technology
Operating System	Linux / WSL (Ubuntu on Windows)
Compiler	GNU C++ (g++) & NVIDIA CUDA (nvcc)
Programming Language	C++
LBM Framework / Library	OpenLB (olb_cpu, olb_gpu)
Hardware / Compute	Dell Laptop (16 GB RAM)
Processors	CPU & GPU
Parallel Computing Frameworks	OpenMP & NVIDIA CUDA Toolkit
Visualization	ParaView

Table 1: Computational tools, libraries, and hardware used for the LBM simulations.

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